

Self-Supervised Learning based Speech Representations

DEEP SPEECH PROCESSING Course

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Module: Self-Supervised Learning based Speech Representations

1. **Representation learning for speech:** a brief historical context.
2. **Self-Supervised Learning (SSL):** Motivation: reducing labeled data requirements, leveraging large unlabeled corpora. General frameworks: contrastive, generative, and predictive.
3. **SSL-based Pretraining:** Mockingjay, wav2vec 2.0, HuBERT, etc.
4. **Fine-tuning and domain adaptation:** Fine-tuning SSL models for downstream tasks, resource-constrained adaptation (low-resource languages, few-shot learning)
5. **Benchmarking:** To understand information encoded in speech representations and to, evaluate and compare their performance. For example, SUPERB, IndicSUPERB, etc.

Key references:

1. *Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, [arXiv:2110.09327](https://arxiv.org/abs/2110.09327)*
2. *Abdelrahman Mohamed, et al., Self-Supervised Speech Representation Learning: A Review, arxiv.2205.10643*
3. *A. Baevski, et al., "wav2vec 2.0: A framework for self-supervised learning of speech representations," Proceedings of NeurIPS, vol. 33, 2020.*
4. *W.-N. Hsu, et al., "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," arXiv preprint arXiv:2106.07447, 2021.*
5. *Javed, Tahir, et al. "Indicsuperb: A speech processing universal performance benchmark for Indian languages." Proceedings of AAAI Conference on Artificial Intelligence. Vol. 37. No. 11. 2023.*
6. *Arora, Siddhant, et al. "On the Evaluation of Speech Foundation Models for Spoken Language Understanding." arXiv preprint arXiv:2406.10083 (2024).*



Yann LeCun's "cake analogy" at [NIPS 2016](#): "If intelligence is a cake, the bulk of the cake is unsupervised learning, the icing on the cake is supervised learning, and the cherry on the cake is reinforcement learning (RL)."

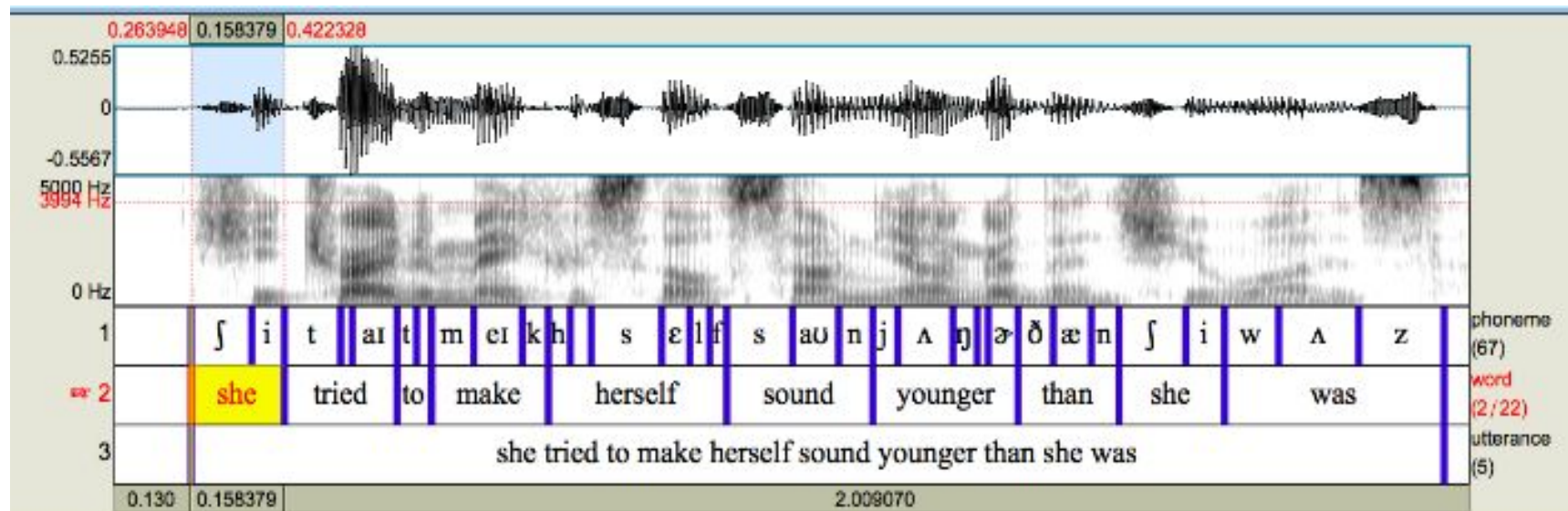
Let's learn 🤗

- Understanding speech signal
- Handcrafted vs learned speech representations
- A brief historical context
- What is self-supervised learning (SSL)?
- Idea of pretraining and finetuning
- Understanding latest SSL-based pretrained models
- Information encoded in the SSL models



Information Encoded in Speech Signals





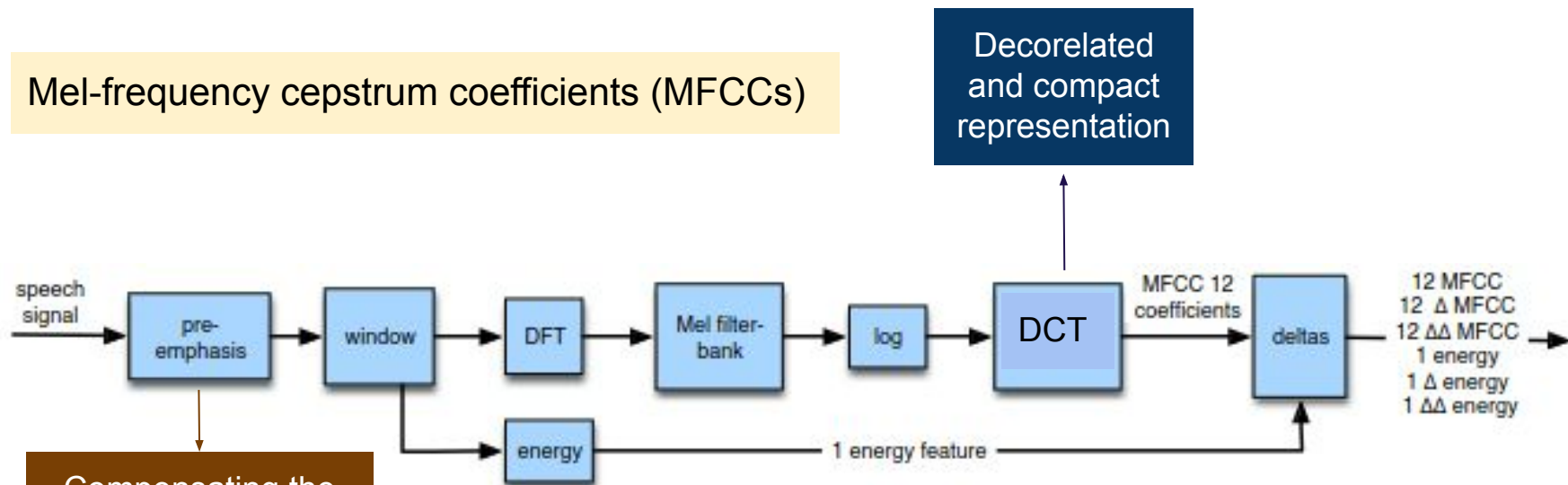
Speech representations

Instead of working directly with high-dimensional, time-domain waveforms (which capture every detail of the signal, including noise and irrelevant variability), we transform them into a feature space that emphasizes the underlying linguistic, phonetic, and speaker-specific information.

Can be derived using feature engineering OR learned from the data

Handcrafted speech features

Mel-frequency cepstrum coefficients (MFCCs)



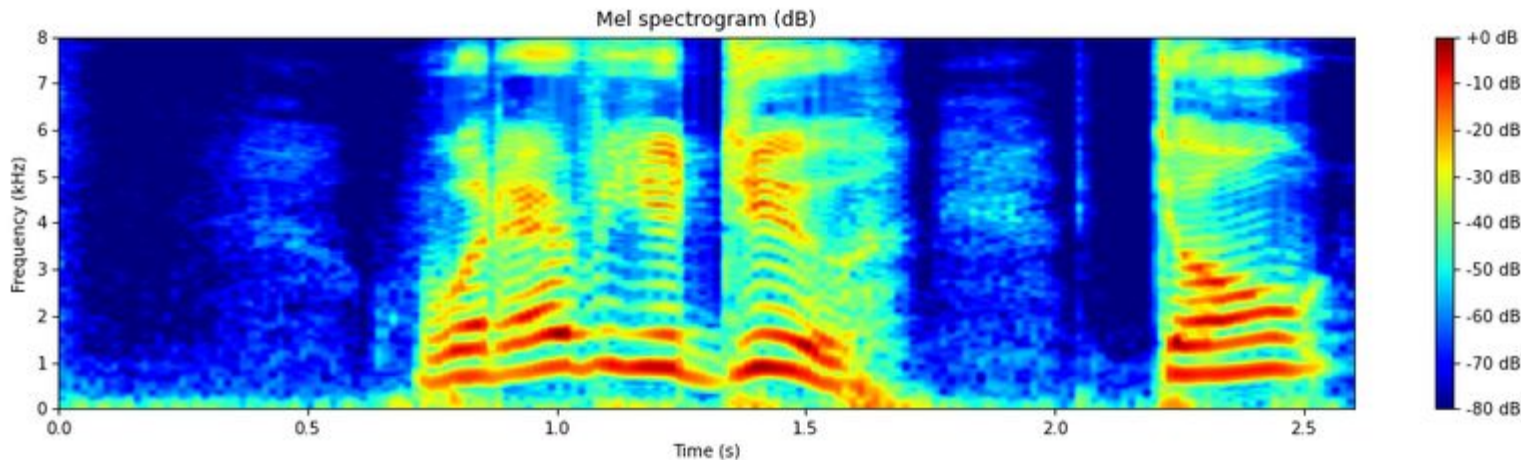
Compensating the rapid decaying spectrum of speech

The Mel-filter bank is inspired by auditory and physiological findings of how humans perceive speech signals

Mel spectrogram

It is found that the last step (DCT) removes the information and destroys spatial relations.

Therefore, it is usually omitted, which yields the log-Mel spectrogram, a popular input for DL models.



Issues with handcrafted speech features

- The feature engineering (based on signal processing, perception analysis) is manual, cumbersome and requires human knowledge
- The designed features might not be best for the objective at hand - MFCCs may not be good for task where we need information about the prosody.

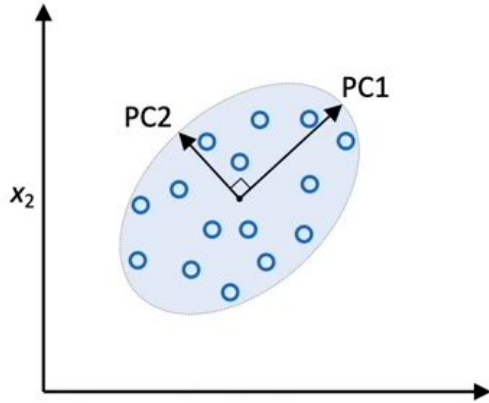
Desirable: To broaden the scope of ML algorithms, it is desirable to make learning algorithms less dependent on hand-crafted features.

Traditional Feature Learning: shallow learning algorithms

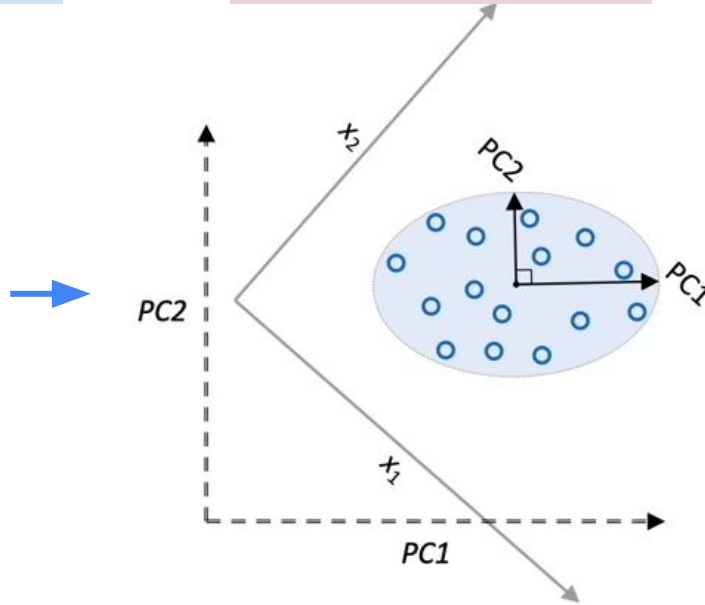
- Principal Components Analysis (PCA),
- Linear Discriminant Analysis (LDA),
- Canonical Correlation Analysis (CCA),
- Non-negative Matrix Factorisation (NMF)

PCA

Find direction of maximum variance



Linear transformation

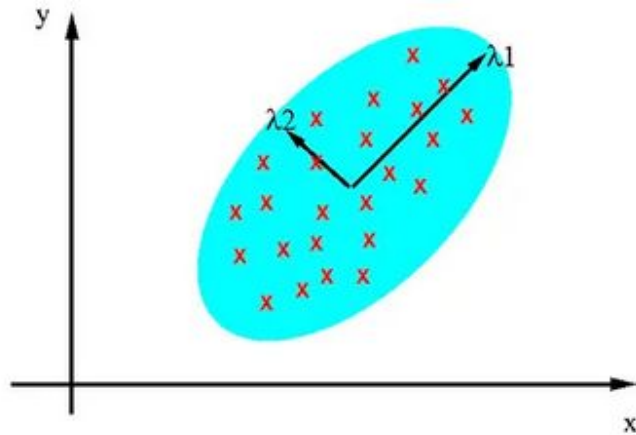


Consider vectors with most variance

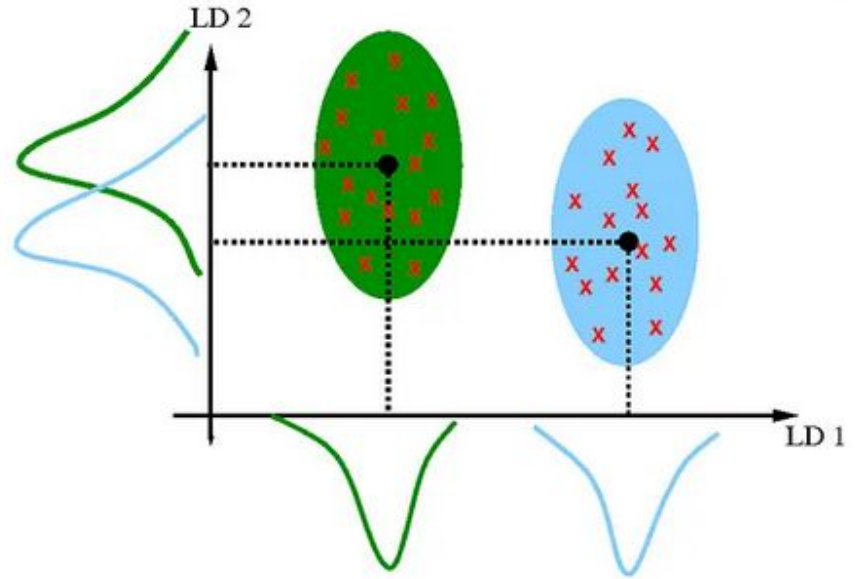


PCA vs LDA

PCA: component axes that maximize the variance



LDA: maximizing the component axes for class-separation



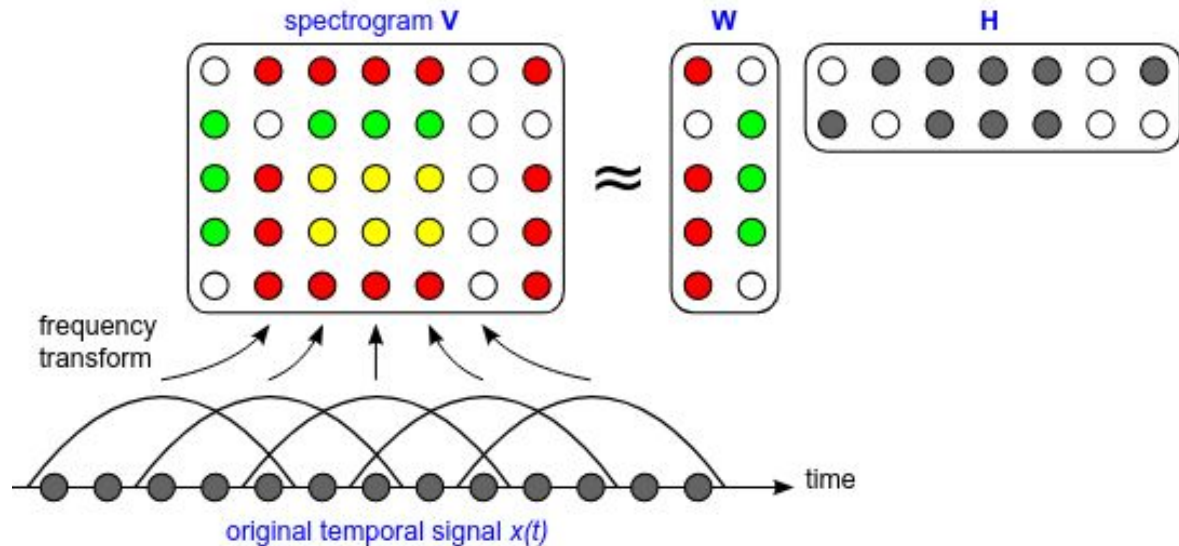
[1] Image source: <https://medium.com/@seshu8hachi/pca-vs-lda-no-more-confusion-fc21fb8d06e9>

[2] <https://sebastianraschka.com/faq/docs/lda-vs-pca.html>

NMF for speech

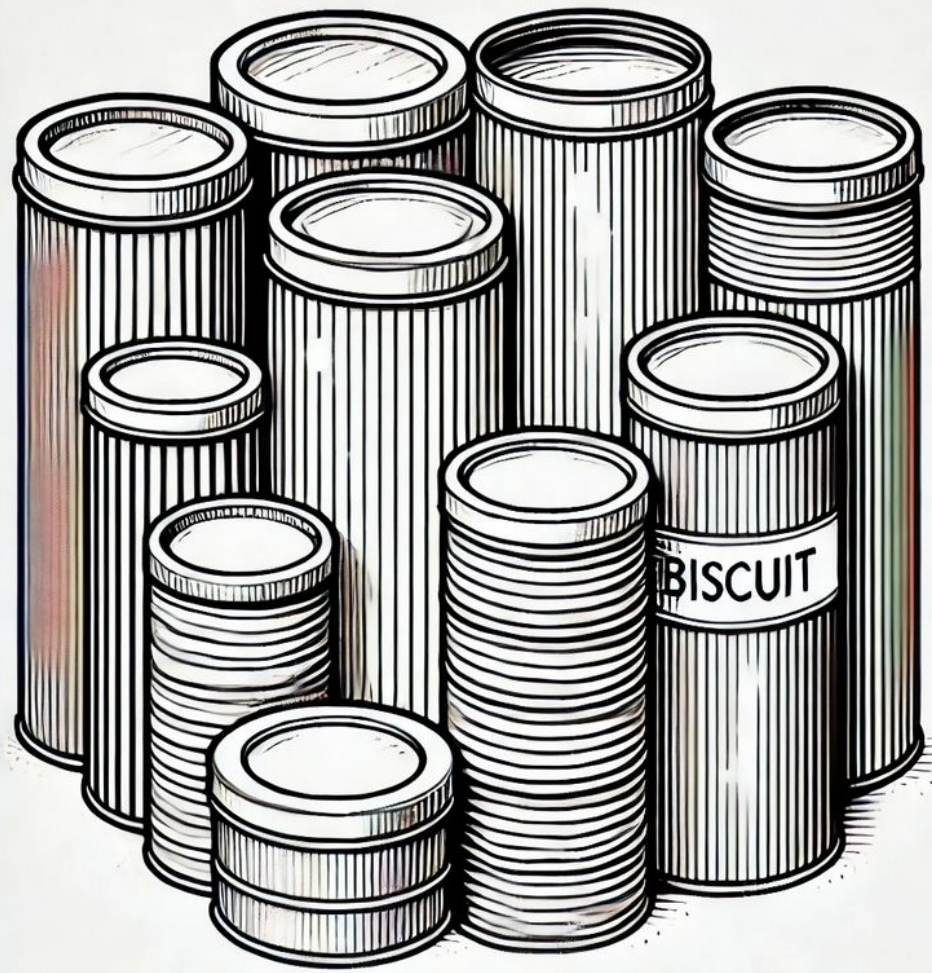
W: Represents the set of "building blocks" (basis)

H: Activation matrix



What is representation learning (RL)?

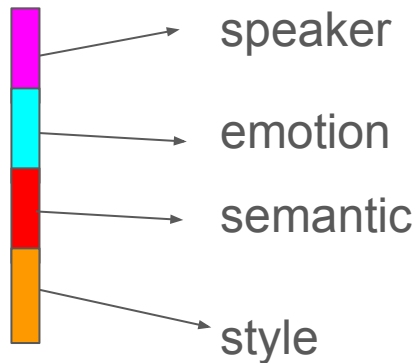
- Discover latent features automatically - providing a richer, more meaningful way to describe and understand the data [1]. “*listens*” to speech and comes up with its own internal summary of that speech.
- Speech representations that span from basic acoustic features up to higher-level lexical, syntactic, and semantic information.
- Unsupervised approach (self-supervised learning is a subset), that discover naturally occurring patterns in training data.



What do we expect?

- **Disentangled**
 - Learn representations that disentangle the speech variations (factor of variations), such as speaker identity, style, emotion.
- **Invariant**
 - Maintain stability despite changes in background noise or communication channels. Ensure robustness for different application scenarios.
- **Hierarchical**
 - Capture features from low-level acoustics up to higher-level semantics.
 - Support diverse tasks (e.g., speaker ID may need a low-level acoustic representation; however, translation may need semantic level information to encode).

Disentangle means separating these factors and encoding them into independent and distinct latent variables in the representation space.



Characteristics of speech as compared to image and text

High-dimensional, long, and variable-length sequences.

Speech is continuous. Therefore, can't define a finite vocabulary.

Complex hierarchical structure that is difficult to infer without supervision - no clear segmentation segment boundaries

Speech processing tasks are diverse. Different speech tasks may require mutually orthogonal information.

Variability: speaker, accent, recording condition, etc

Data-driven speech representation

Clustering and
Mixture Models

Stacked Neural
Models

Pretraining-based
using SSL

Clustering based speech representations

- It aims to categorise similar classes of data samples into one cluster using similarity measures (e. g., Euclidean distance).
- Word patterns are clustered semi automatically using techniques such as k-means - after which isolated words are recognized by finding the training cluster closest to the test data [1].

GMM

GMM facilitated the modeling of more variability in the input data.

Stack mean vectors for speech representation

The probability given in a mixture of K Gaussians is:

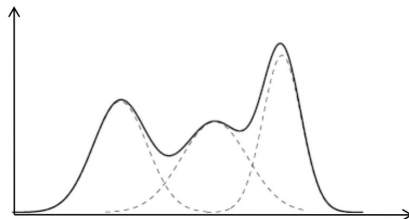
$$p(x) = \sum_{j=1}^K w_j \cdot N(x | \mu_j, \Sigma_j)$$

where w_j is the prior probability (weight) of the j th Gaussian.

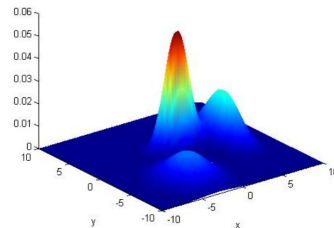
$$\sum_{j=1}^K w_j = 1 \quad \text{and} \quad 0 \leq w_j \leq 1$$

Examples:

d=1:

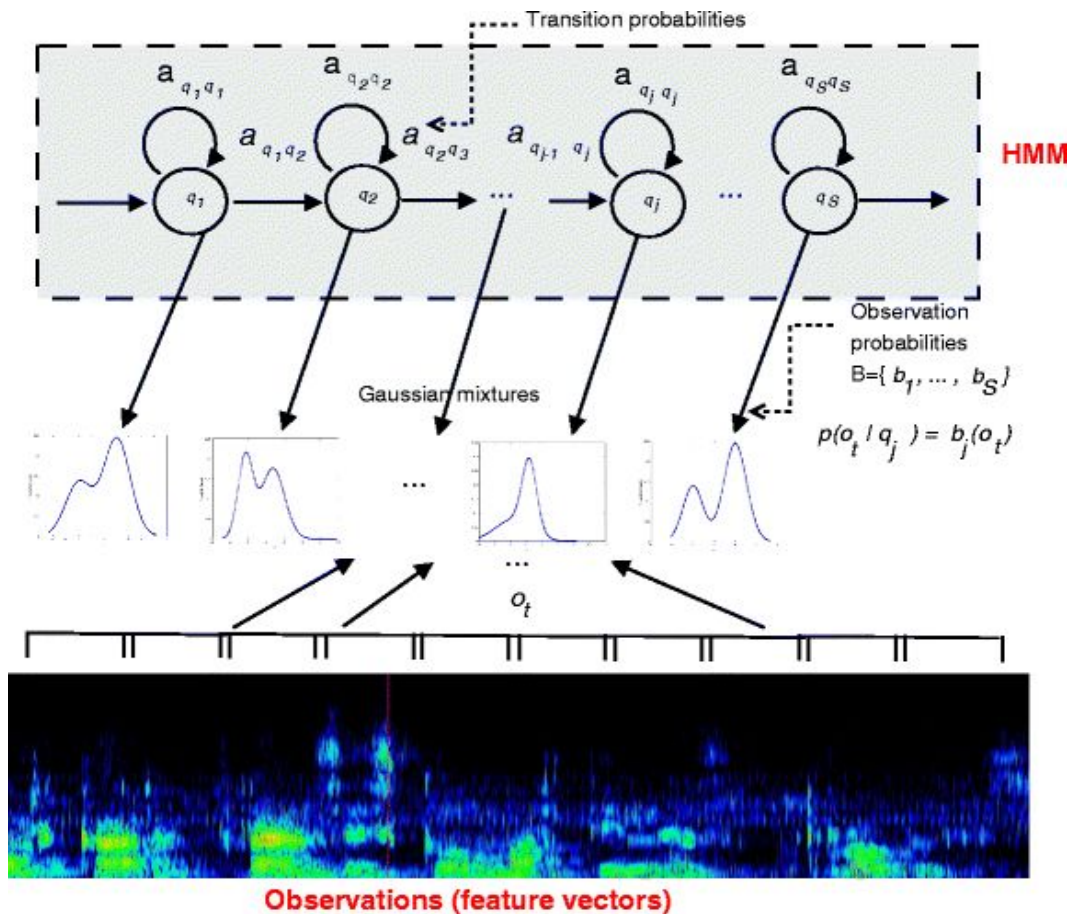


d=2:



HMM

HMM model temporal dynamics in speech, whereas the GMM clusters features without temporal context.



Neural models

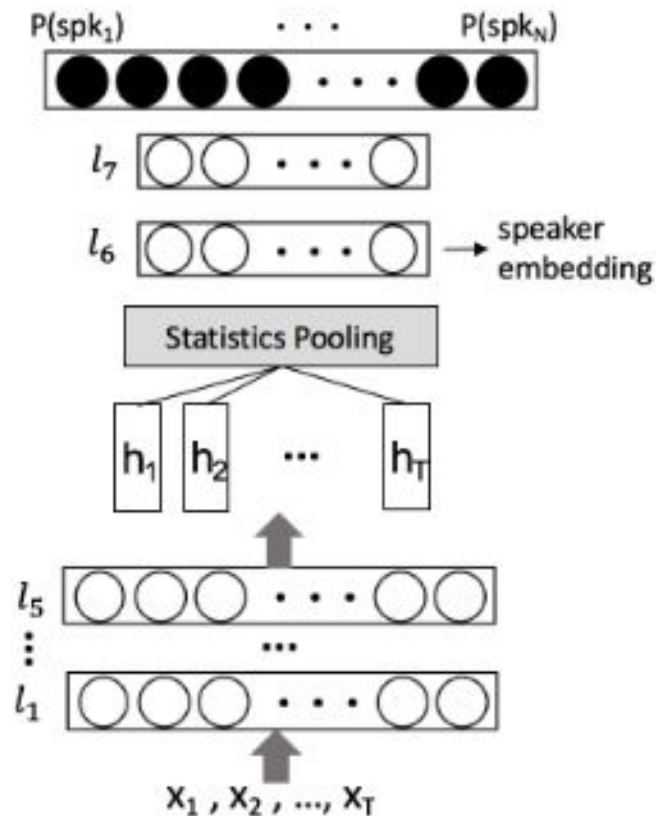
- Autoencoders
- DNN
- RNN
- CNN
- Transformer

DNN

For speaker recognition – a DNN is trained to classify speakers, and the averaged outputs of its last hidden layer form an embedding for each speaker

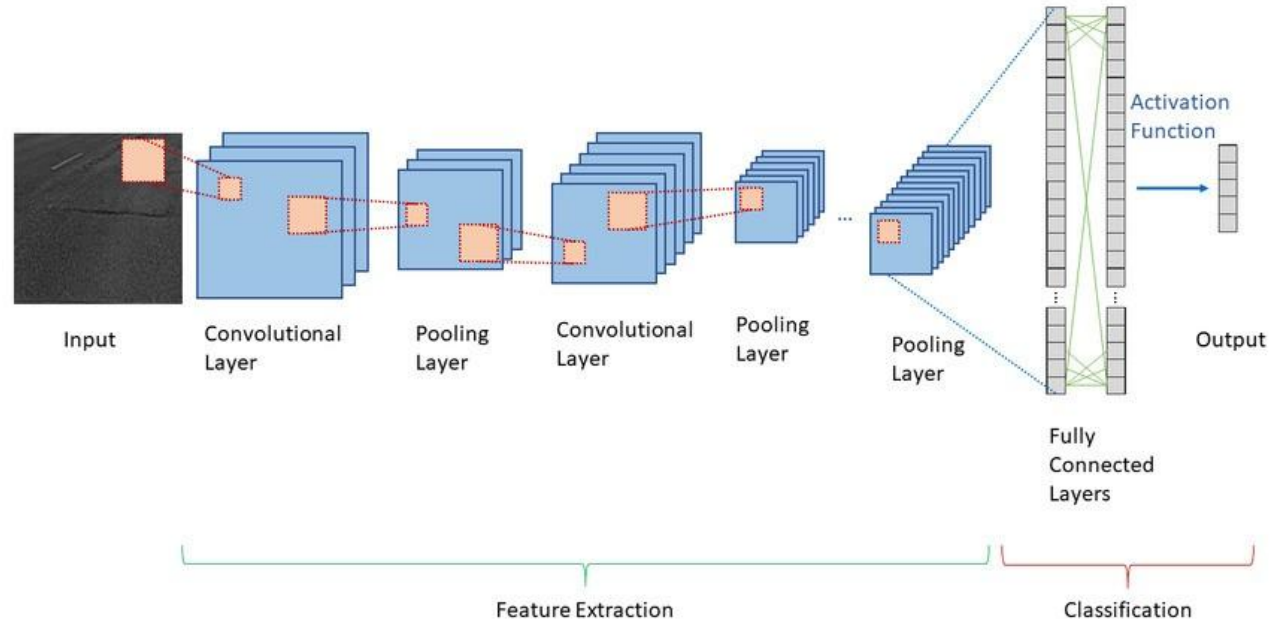
Embeddings are **compact and discriminative**: they encapsulate the speaker's voice characteristics.

Can be extended to language identification

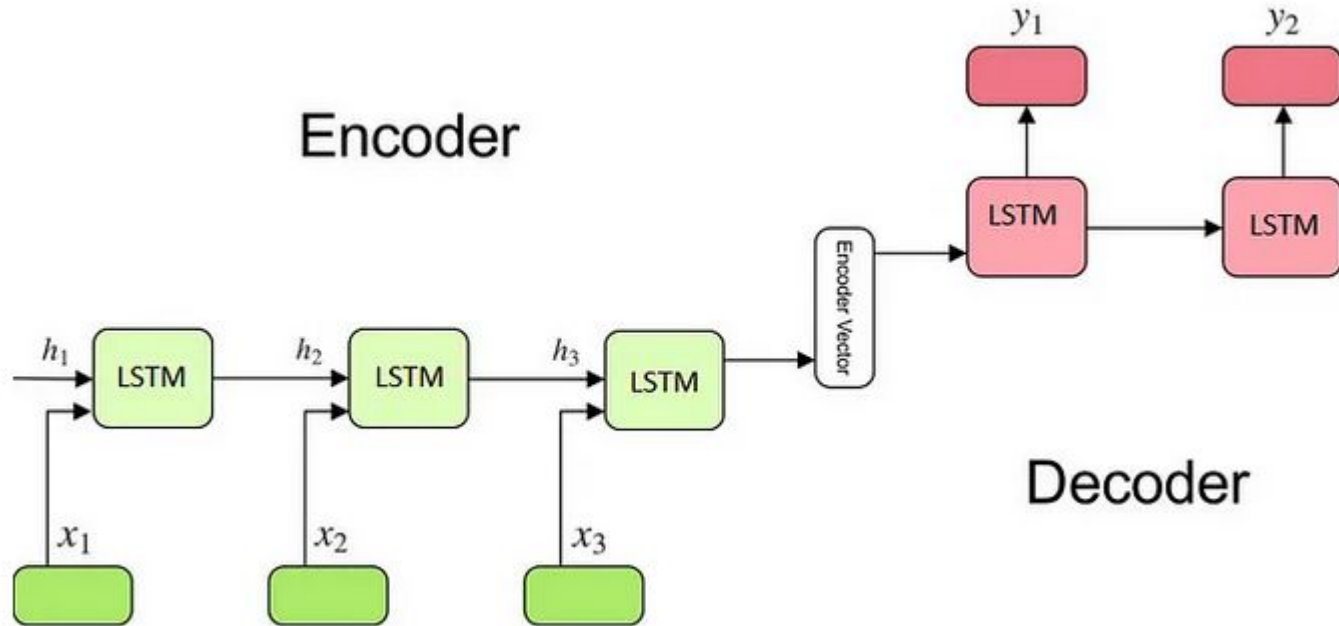


Convolutional Neural Networks (CNNs)

CNN's filters act like automatic feature extractors, scanning the speech spectrogram (raw speech with 1D CNN) for modeling distinctive patterns



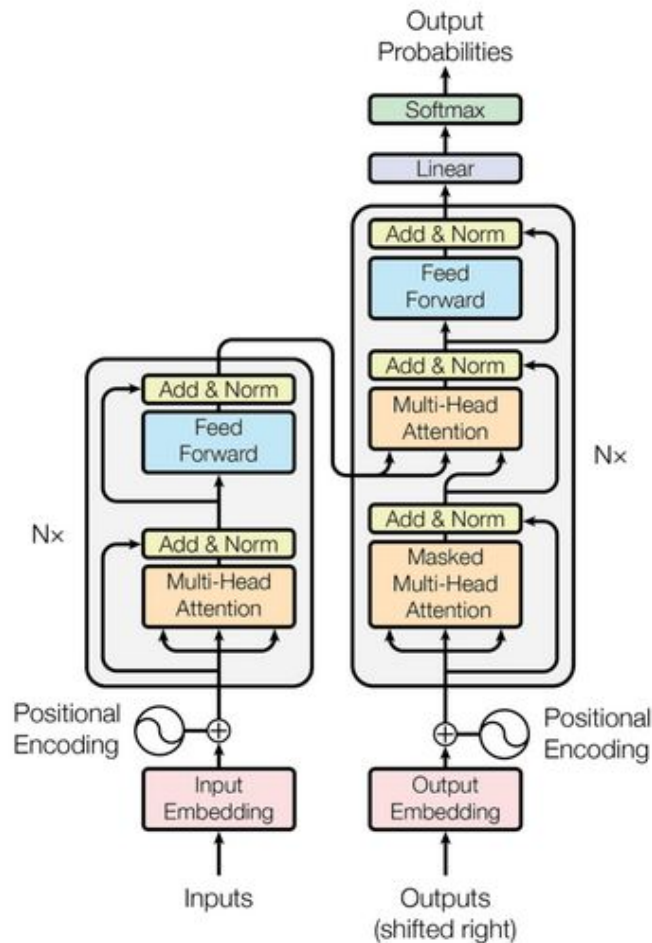
Recurrent neural networks (RNNs)



Transformers

Transformers are a type of neural network architecture that uses self-attention mechanisms to process sequences of data. They can learn relationships between words in a sentence

Transformer: main component of most of the recent SSL models



The encoder-decoder structure of the Transformer architecture
Taken from "Attention Is All You Need"

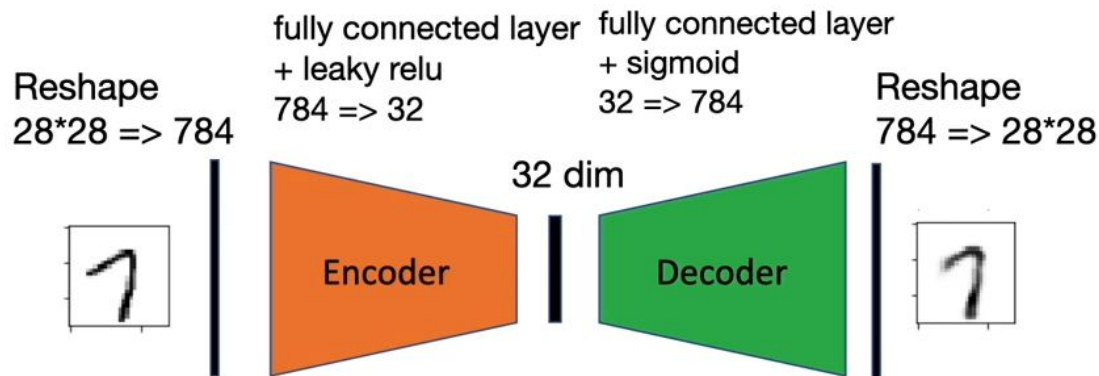
Autoencoders: a type of unsupervised learning

Encoder: `torch.nn.Linear(784, 32)`

Decoder: `torch.nn.Linear(32, 784)`

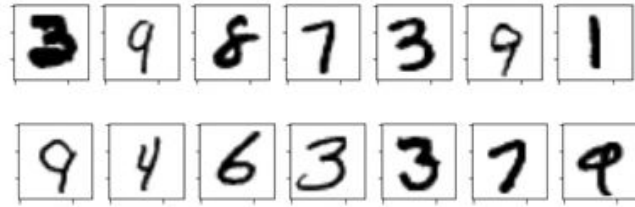
$$MSE(x, \hat{x}) = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

No orthogonality constraint
as PCA for latent
dimension

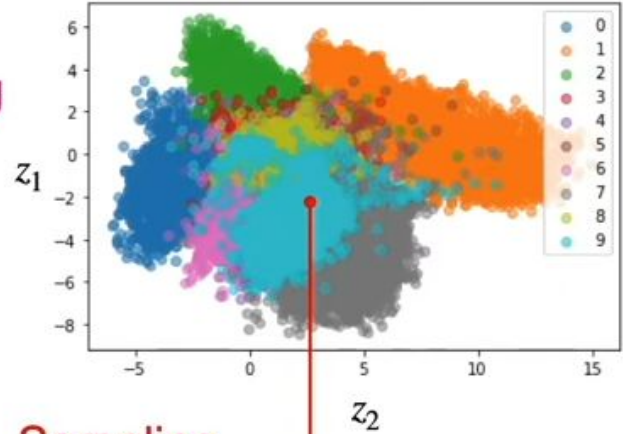


In essence, it's learning "what features of the speech can I not afford to lose?"

Autoencoder contd.



Encoding



Sampling
& Decoding



Autoencoder contd.

Deterministic: The encoder directly outputs a fixed point in the latent space, without any probabilistic interpretation.

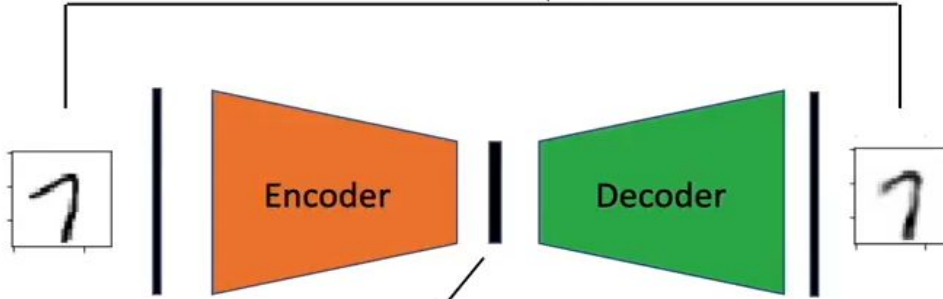
- representing each data point as a single point in the latent space

Limited generation: Due to the deterministic nature, generating new data points is more challenging as there is no mechanism to introduce variation

Variational autoencoders (VAE)

1) Minimize squared error loss: (ensures good reconstruction)

$$\mathcal{L}_1 = \|\mathbf{x} - \text{Dec}(\text{Enc}(\mathbf{x}))\|_2^2 = \sqrt{\sum_{i=1}^d (x_i - x'_i)^2}$$



It encodes each input as a *distribution* of possible latent variables

2) Minimize KL divergence:

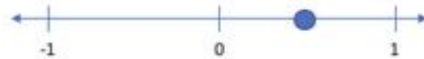
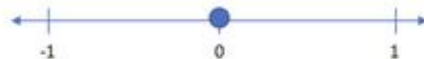
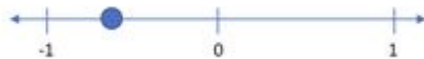
(ensures latent space is continuous and standard normal distributed)

$$\mathcal{L}_2 = D_{KL}[N(\mu, \sigma) \| N(0, 1)] = -\frac{1}{2} \sum (1 + \log(\sigma^2) - \mu^2 - \sigma^2)$$

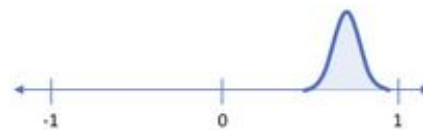
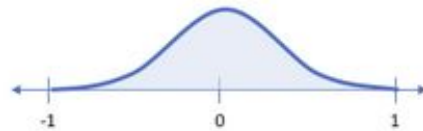
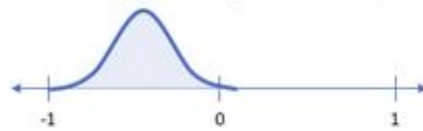
Overall loss: $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2$



Smile (discrete value)

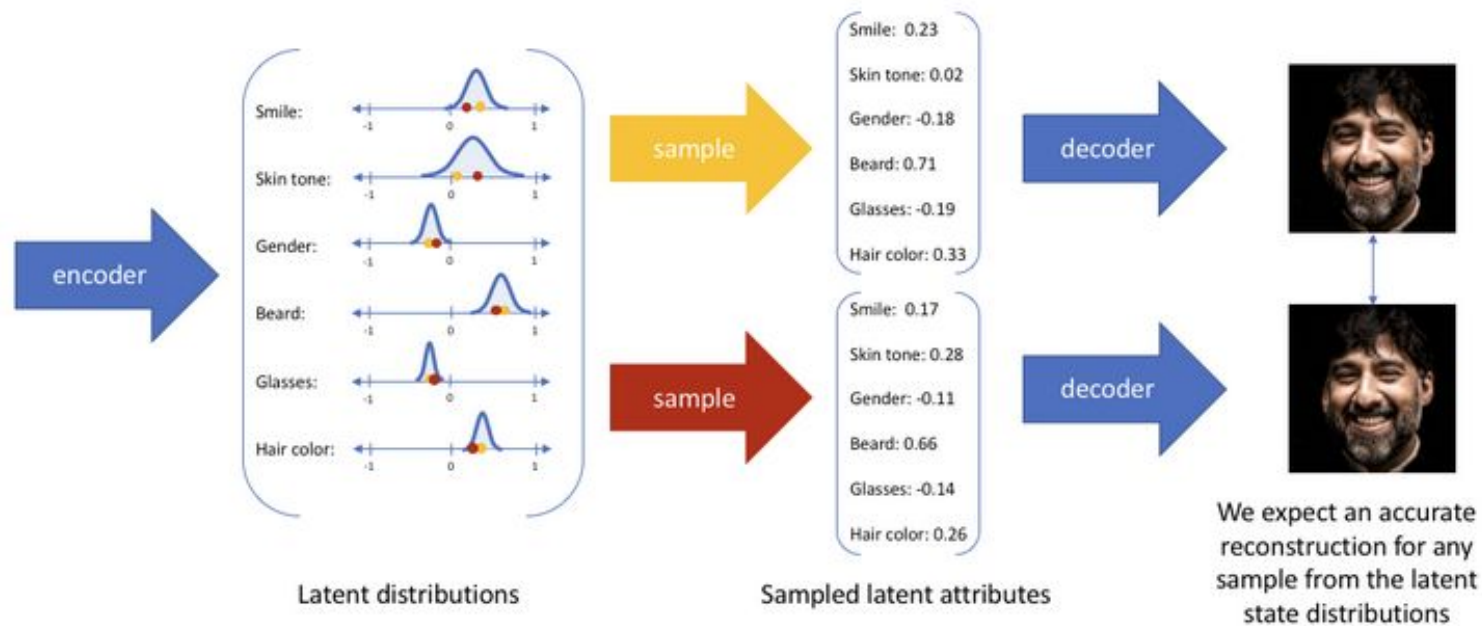


Smile (probability distribution)



vs.

Sampling from the distribution



Variational autoencoders (VAE)

Probabilistic: The encoder outputs the parameters (mean and variance) of a probability distribution, allowing for sampling from the latent space to generate new data points.

Continuous: The latent space is usually continuous, enabling smooth transitions between data points by manipulating the latent variables.

KL Divergence: It is used to encourage the encoded distribution to resemble a standard normal distribution.